

COURSEPACK

SCHEME

The scheme is an overview of work-integrated learning opportunities and gets students out into the real world. This will give what a course entails.

Course Title	Engineering Mathematics-II			Course Type		Integrated			
Course Code				Class		B. Tech 2 nd Sem			
Instruction delivery	Activity	Credits	Credit Hours	Total Number of Classes per Semester				Assessment in Weightage	
	Lecture	3	3	Theory	Tutorial	Practical	Self-study	CIE	SEE
	Tutorial	0	0						
	Practical	1	2						
	Self-study	0	0						
	Total	4	5	45	0	15	0	50%	50%
Course Lead			Course Coordinator						
Names Course Instructors	Theory			Practical					

COURSE OVERVIEW: This course is familiarizing the prospective engineers with techniques in Engineering Mathematics-I. It aims to equip the students with standard concepts and tools at an intermediate to advance level that will serve them well towards tackling more advanced level of Mathematics and application that they would find useful in their discipline.

PREREQUISITE COURSE

PREREQUISITE COURSE REQUIRED	YES /NO	
If, yes please fill in the Details	Prerequisite course code	Prerequisite course name
	C1UC122B	Engineering Mathematics-I

COURSE OBJECTIVE:

The students will be able:

1. to visualize and conceptualize the engineering problems.
2. to model the engineering problem mathematically using theory of calculus and matrices.
3. to determine the solution of the studied engineering problems from application point of view.
4. to validate the solution.
5. to implement the solution for engineering problem

COURSE OUTCOMES (COs)

After the completion of the course, the student will be able to:

CO No.	Course Outcomes
CO1	Summarize the concepts of vector calculus, vector space, Inner product spaces, and differential equations.
CO2	Explain vector differentiation & integration, properties of Vector space, Linear maps and methods for solving Differential equations.
CO3	Apply the concept of rank-nullity theorem, Gram-Schmidt orthogonalization, and appropriate methods to solve differential equations.
CO4	Analyze the concept of Green's, Stoke's and Gauss-divergence theorem, Ordinary and partial differential equations to solve various real world problems.

BLOOM'S LEVEL OF THE COURSE OUTCOMES

Bloom's taxonomy is a set of hierarchical models used for the classification of educational learning objectives into levels of complexity and specificity. The learning domains are cognitive, affective, and psychomotor.

THEORY

CO No.	Remember BTL1	Understand BTL2	Apply BTL3	Analyse BTL4	Evaluate BTL2	Create BTL6
1	.					
2		.				
3			.			
4			.			

PROGRAM OUTCOMES (POs): AS DEFINED BY CONCERNED THE APEX BODIES

PO1	Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO2	Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO3	Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
PO4	Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO5	Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO6	The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues, and the consequent responsibilities relevant to the professional engineering practice.
PO7	Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, demonstrate the knowledge of, and need for sustainable development.
PO8	Ethics: Apply ethical principles and commit to professional ethics, responsibilities, and norms of the engineering practice.
PO9	Individual and Teamwork: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO10	Communication: Communicate effectively on complex engineering activities with the engineering community and with society, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO11	Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO12	Life-long learning: Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PSO1: Ability to work with emerging technologies in computing requisite to Industry 4.0

PSO2: Demonstrate Engineering Practice learned through industry internship to solve live problems in various domains.

COURSE ARTICULATION MATRIX

The Course articulation matrix indicates the correlation between Course Outcomes and Program Outcomes and their expected strength of mapping in three levels (low, medium, and high).

COs#/ POs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO 2
CO204T1	3	3	1	1									2	2
CO204T2	3	3	1	1									2	1
CO204T3	3	2	1	1									1	1
CO204T4	3	2	1	1									2	2

Note: 1-Low, 2-Medium, 3-High

COURSE ASSESSMENT

Assessment pattern for Blended/ Integrated course

Type of Course	CIE Weightage			End Term Exam (ETE) Weightage
	LAB (Daily Work/ Record)	LAB EXAM	Mid Term Exam	
Integrated (B)	25	25	50	50
Final Weightage	25		25	50
Total	100			

COURSE

Content
Scalar and vector fields, Differentiation of Vector functions, Gradient, divergence, curl, line integrals, path independence, potential functions and conservative fields, surface integrals, Green's theorem, Stokes's theorem and Gauss's divergence theorems (without proof & simple problems). Vector Space, Linear Independence of vectors, basis, dimension; Linear transformations (maps), range and kernel of a linear map, rank, nullity, rank-nullity theorem, Inverse of a linear transformation, composition of linear maps, Matrix associated with a linear map.

Inner product spaces, Norms, Orthogonality, Orthogonal and Orthonormal Basis, Orthogonal Projections, Gram-Schmidt orthogonalization.

Basic concepts, Exact differential equations, Linear differential equations of second and higher order with constant coefficients, Method of variation of parameters, Cauchy-Euler equation, System of linear differential equations with constant coefficients, applications of linear differential equations.

Basic concepts, Classification of second order linear PDE, Method of separation of variables and its application in solving Wave equation (one dimension), heat equation (one dimension) and Laplace equation (two-dimension steady state only).

BIBLIOGRAPHY

Text Books:

1. D. Poole, Linear Algebra: A Modern Introduction, 2nd Edition, Brooks/Cole, 2005.
2. Erwin Kreyszig, Advanced Engineering Mathematics, 10th Edition, John Wiley & Sons.
3. R. K. Jain and S. R. K. Iyengar, Advanced Engineering Mathematics, 4th Edition, Narosa Publishers.
4. Robert T. Smith and Roland B. Minton, Calculus, 4th Edition, McGraw Hill Education.
5. George B. Thomas and Ross L. Finney, Calculus, 9th Edition, Pearson Education.

Reference Books:

1. Robert T. Smith and Roland B. Minton, Calculus, 4th Edition, McGraw Hill Education.
2. David C Lay, Linear Algebra and its application, 3rd Edition, Pearson Education.
3. Michael D. Greenberg, Advanced Engineering Mathematics, 2nd Edition, Pearson Education.

Course Name: Exploration with CAS-II

Objective: *The objective of this course is to enhance the problem-solving skills of prospective engineers using open-source software/ computer algebra system and to perform tedious and difficult algebraic manipulations/tasks as well as plotting of graphs for complicated functions to understand their behavior.*

Lab-1:

Revision of the Scilab: Overview, Basic syntax, Mathematical Operators, Predefined constants, Built in functions. Conditional Statements, Loops, Matrix, and Its applications.

Lab-2:

Basic Vector Calculations, Finding Norm, Angle between two Vectors, Unit Vector.

Lab 3:

Finding Divergence using Scilab and verifying Divergence Theorem, Finding Curl using Scilab and Verifying Stokes Theorem.

Lab-4:

Checking LI and LD of Vectors.

Lab-5:

Verifying Rank-Nullity Theorem.

Lab-6:

Orthogonal and Orthonormal Vectors, Gramm Schmidt Orthogonalization process.

Lab-7:

Matrix associated with linear Transformations and their corresponding operations.

Lab-8:

Plotting Direction fields of first order differential equations.

Lab-9:

Solving Linear ODE with different initial conditions using Scilab.

Lab-10:

Solving PDE with boundary conditions using scilab.

Textbooks (For Tutorial sessions):

1. Robert T. Smith and Roland B. Minton, *Calculus*, 4th Edition, McGraw Hill Education.
2. George B. Thomas and Ross L. Finney, *Calculus*, 9th Edition, Pearson Education

References for Lab sessions (On scilab):

a. Urroz, G E., *Numerical and Statistical Methods with SCILAB for Science and Engineering*, Vol 1 BookSurge Publishing, 2001, ISBN-13: 978-1588983046

a. Software site: <http://www.scilab.org>, official scilab website

Wikipedia article: <http://en.wikipedia.org/wiki/Scilab>

LESSON PLAN FOR Integrated COURSES

FOR THEORY 15 weeks * 3 Hours = 45 Classes) (1credit = 1Lecture Hour)

FOR PRACTICAL 15 weeks * 2Hours = 30 Hours lab sessions (1 credit = 2 lab hours)

L-N o	Topic for Delivery	Tutorial/ Practical Plan	Skills	Competency
1	Scalar and vector fields, Differentiation of Vector functions	Theory	Ability to utilize the concepts of vector and scalar fields	Understanding of fundamental vector calculus concepts and applications.
2	Formula for finding gradient, Geometrical meaning as normal vector, Formulas to find divergence and curl of vector fields, Illustrative examples	Theory		
3	Defining Line integrals as the limit of a finite sum and work done by a force field, Methods to evaluate Line integrals, Illustrative examples	Theory		
4	Revision of the Scilab: Overview, Basic syntax,	Practical		
5	Mathematical Operators, Predefined constants, Built in functions. Conditional Statements, Loops, Matrix, and Its applications.	Practical		
6	Statement of Green's Theorem, Application to evaluate Line integrals, Evaluating plane area, Illustrative examples	Theory		
7	Statement of Stokes's theorem, Based numericals	Theory		
8	Statement of Gauss's theorem, Application to find surface integral over Cubical and spherical surfaces only, Jacobian with simple problems	Theory		

9	Basic Vector Calculations, Finding Norm, Angle between two Vectors, Unit Vector.	Practical		
10		Practical		
11	Statement of Gauss's theorem, Application to find surface integral over Cubical and spherical surfaces only, Jacobian with simple problems.	Theory	Ability to use the concept of vector space, linear transformation and Inner product space	demonstrating a deep understanding of linear algebra and its applications.
12	Definition of vector space, Linear independence and dependence of vectors, basis and dimension	Theory		
13	Linear transformations, Range and Kernel	Theory		
14	Finding Divergence using Scilab and verifying Divergence Theorem, Finding Curl using Scilab and Verifying Stokes Theorem.	Practical		
15		Practical		
16	Rank and nullity of Linear transformation	Theory		
17	rank- nullity theorem (only statement) and its applications	Theory		
18	Composition of Linear map, Inverse of Linear Transformation	Theory		
19	Checking LI and LD of Vectors.	Practical		
20		Practical		
21	Matrix associated with linear map	Theory		
22	Revision of definition of field, and define inner product space (for complex and real field both) with examples	Theory		
23	Revision of definition of field, and define inner product space (for complex and real field both) with examples.	Theory		
24	Verifying Rank-Nullity Theorem.	Practical		
25		Practical		
26	Define orthogonal sets and orthonormal sets with examples.	Theory		
27	Define Gram-Schmidt orthogonalizations process and Solve problems related to its application.	Theory		
28	Define Gram-Schmidt orthogonalizations process and Solve problems related to its application.	Theory		
29	Matrix associated with linear Transformations and their corresponding operations.	Practical		
30		Practical		
31	Defining first order exact differential equation, necessary and sufficient condition, General solution.	Theory	ability to analyze solution of differential equation and their integral curves	understanding of foundational concepts in differential equation
32	n th order homogeneous linear equation $f(D)y=0$, linear independence of solutions,	Theory		
33	auxiliary equation, solution: when roots of auxiliary equation are a) distinct, b) equal, c) complex.	Theory		
34	Orthogonal and Orthonormal Vectors, Gramm Schmidt Orthogonalization process.	Practical		
35		Practical		
36	n th order non-homogeneous linear equation $f(D)y=r(x)$,	Theory		
37	general solution =Complimentary function+ particular integral, method to find PI when $r(x)=e^{ax}$.	Theory		

38	Method to find PI when $r(x) = \sin(ax)$, $\cos(ax)$. And $r(x) = x^n$.	Theory		
39	Plotting Direction fields of first order differential equations.	Practical		
40		Practical		
41	Method to find PI when $r(x) = \sin(ax)$, $\cos(ax)$. And $r(x) = x^n$.	Theory		
42	method to find PI when $r(x) = \exp(ax)V(x)$, $x^n \sin(ax)$, $x^n \cos(ax)$	Theory		
43	method to find PI when $r(x) = \exp(ax)V(x)$, $x^n \sin(ax)$, $x^n \cos(ax)$	Theory		
44	Solving Linear ODE with different initial conditions using	Practical		
45	Scilab.	Practical		
46	Variation of parameter method to find PI of a second order linear differential equation.	Theory		
47	Variation of parameter method to find PI of a second order linear differential equation.	Theory		
48	Cauchy-Euler equation and its solution.	Theory		
49	Solving Linear ODE with different initial conditions	Practical		
50	using Scilab.	Practical		
51	Cauchy-Euler equation and its solution.	Theory		
52	Finding solution of a system of linear equations.	Theory		
53	Finding solution of a system of linear equations.	Theory		
54	Solving Linear ODE with different initial conditions using	Practical		
55	Scilab.	Practical		
56	Mathematical modeling. Basic elements of an electric circuit, Kirchhoff's law.	Theory	Ability to solving the mathematical problems involving differential equations, and numerous applications in physics, engineering, economics	utilizing Heat, Wave and Laplace equation to solve various real-world problems,
57	Solution of simple LR and CR circuits.	Theory		
58	Basic concept and classification of second order PDE	Theory		
59	Solving PDE with boundary conditions using scilab.	Practical		
60		Practical		
61	Separation of variable method to solve second orders linear homogeneous PDEs with constant coefficients	Theory		
62	One dimensional wave equation as mathematical model of vibrations of a stretched string. Solution of 1-dim wave equation by SOV method.	Theory		
63	Solution of 1-dim wave equation with different initial conditions.	Theory		
64	Solving PDE with boundary conditions using scilab	Practical		
65		Practical		

			, and other fields.	
66	One dimensional heat equation as mathematical model for the temperature distribution in a thin heated rod.	Theory		
67	Solution of 1-dim heat equation with both ends of the rod at infinity.	Theory		
68	Solution of 1-dim heat equation when the rod has insulated ends.	Theory		
69	Revision of finding Curl using Scilab and Verifying Stokes Theorem.	Practical		
70		Practical		
71	Two-dimensional Laplace equation.	Theory		
72	Two-dimensional Laplace equation as a mathematical model for the steady state temperature distribution in a thin rectangular plate, solution of the equation.	Theory		
73	Solution of Laplace equation with different boundary conditions.	Theory		
74	Revision of finding Divergence using Scilab and verifying Divergence Theorem, Finding Curl using Scilab and Verifying Stokes Theorem.	Practical		
75		Practical		

PROBLEM-BASED LEARNING

Exercises in Problem-based Learning (Assignments)

Sr. No.	Problem	KL
1	Define gradient of the scalar field with example.	K2
2	Find the gradient of the scalar field $f(x, y, z) = x^2 y^2 + xy^2 - z^2$.	
3	If $r = x i + y j + z k$, $ r = r$ and $\hat{r} = \frac{r}{r}$, then show that $\text{grad}\left(\frac{1}{r}\right) = -\frac{\hat{r}}{r^2}$ where $\hat{r}$ is unit vector.	K3
4	Let $v = a(x, y, z) i + b(x, y, z) j - c(x, y, z) k$ be a differentiable vector field. Show that $\text{div}(\text{curl } v) = 0$.	K3
5	Let $f(x, y, z) = 16xy^3z^2$ be a differentiable scalar field. Show that $\text{curl}(\text{grad } f) = 0$.	K2, K3
6	Evaluate $\int_C (x^2 + yz) ds$, where C is the curve defined by $x = 4y$, $z = 3$ from $(2, \frac{1}{2}, 3)$ to $(4, 1, 3)$.	K3, K4
7	Evaluate $\oint_C (x^2 + y^2) dx + (y + 2x) dy$, where C is the boundary of the region in the first quadrant that is bounded by the curves $y^2 = x$, $x^2 = y$ by Green's theorem.	K4, K5
8	Evaluate the surface integral $\iint_S F \cdot n dA$ where $F = z^2 i + xy j - y^2 k$ and S is the portion of the surface of the cylinder $x^2 + y^2 = 36$, $0 \leq z \leq 4$ included in the first octant.	K4, K5
9	Use the Gauss theorem to evaluate $\iint_S (v \cdot n) dA$, where $v = x i + y j + z k$ and S is the boundary of sphere $x^2 + y^2 + z^2 = 4$.	K5
10	Evaluate $\oint_C v \cdot dr$ using the Stokes's theorem where $v = 3y i + 4z j + 2x k$ and C is the intersection of the surface of the sphere $x^2 + y^2 + z^2 = 16$, $x \geq 0$ and the cylinder $y^2 + z^2 = 4$.	K5
11	Discuss whether or not R^2 is a subspace of R^4 	K4
12	Discuss that set of all square matrices of order n form a vector space with respect to matrix addition and scalar multiplication.	K3
13	Determine whether, the set $\{x, x, \cos \cos 2x\}$ is linearly dependent.	K3
14	Find the coordinate vector $[p(x)]$ of $p(x) = 1 - 4x + 6x^2$ with respect to the basis $\{1 + x, x + x^2, 1 + x^2\}$.	K3
15	If $A = [1 \ 1 \ 1 \ 2 \ 2 \ 3 \ x \ y \ z]$ and $V = \{(x, y, z) \in R^3; \det(A) = 0\}$ then find dimension of V ?	K2

16	Let the linear transformation $T: R^3 \rightarrow R^3$ be defined by $T(x, y, z) = (2x + z, 3x - z, x + y + z)$. Then find range space and null space of T .	K3
17	Let the linear transformation $T: R^3 \rightarrow R^3$ be defined by $T(x, y, z) = (x + z, 2x + y + 3z, 2y + 2z)$. Then find matrix of linear transformation T with respect to the ordered basis $B = \{(1, 1, 0), (-1, 0, 1), (1, -2, 3)\}$.	K3
18	Let $T_1, T_2: R^5 \rightarrow R^3$ be the linear transformation such that $rank(T_1) = 3$ and $nullity(T_2) = 3$. Let $T_3: R^3 \rightarrow R^3$ be the linear transformation such that $T_3 \circ T_1 = T_2$. Then find rank of T_3 ?	K2
19	Let the linear transformation $T: p_3[0, 1] \rightarrow p_2[0, 1]$ be defined by $(Tp)(x) = p''(x) + p'(x)$. Then find the matrix representation of T with respect to the basis $\{1, x, x^2, x^3\}$ and $\{1, x, x^2\}$ of $p_3[0, 1]$ and $p_2[0, 1]$ respectively?	K3
20	Prove that the $T: R^3 \rightarrow R^3$ defined by $T(x, y, z) = (x + z, x - z, y)$ is invertible. Find inverse of T .	K2
21	Let $u = (a, b)$ and $v = (c, d)$ be two vectors in R^2 . Show that $\langle u, v \rangle = (ac - bd)$ is not an inner product.	K2
22	Find an orthogonal basis for the subspace W of R^3 given by $W = \{[x \ y \ z]: x - y + 2z = 0\}$.	K2
23	Determine the matrix $[10 \ 20 \ 6 \ 10 \ -20 \ 6 \ -10 \ 0 \ 12]$ is orthogonal. If it is, find its inverse.	K3
24	Find an orthogonal basis for R^3 that contains the vector $x_1 = [1 \ 2 \ 3]$. (by Gram-Schmidt Process)	K2, K3
25	Identify order and degree of the following differential equations $\frac{d^2y}{dx^2} - 3x\sqrt{\frac{dy}{dx}} \frac{dy}{dx} = x$.	K2
26	Find the solution of differential equation $\left \frac{dy}{dx}\right + y = 0$.	K2
27	For the initial value problem $\frac{dy}{dx} = \sin \sin x$, $y(0) = 0$. Find the value of y at $x = \pi/3$.	K2
28	Find the general solution of second order linear homogeneous differential equation $\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 25y = 0$.	K2
29	Determine whether the differential equation $e^{x^2}(2xy \, dx + dy) = 0$ is exact. If exact, solve it.	K2
30	Find the particular solution of the differential equation $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 3y = 2x - 3x^2$.	K2, K3
31	Solve $\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 4y = 10e^{-3x}$.	K3
32	Solve	K3

	$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = 2x e^{-3x} \cos \cos x.$	
33	Solve the non-homogeneous ODE by method of variation of parameter $\frac{d^2y}{dx^2} + y = \sec \sec x.$	K2
34	Find the solution of the differential equation $x^2 \frac{d^2y}{dx^2} + 5x \frac{dy}{dx} + 4y = 0.$	K3
35	Find the current $I(t)$ in the LC – circuit with the following data assuming initial current and charge: $L = 2$ Henry, $C = 0.005$ Farad and $E = 220 \sin 4t$ volts.	K3
36	Solve: $(x^2 D^2 - xD + 1)y = \left(\frac{\log \log x}{x}\right)^2.$	K3
37	Solve the set of simultaneous differential equations $(3D + 1)x + 3Dy = 3t + 1$ $(D - 3)x + Dy = 2t.$	K2
38	Classify the partial differential equation $(x^2 - y^2)u_{xx} + 2(x^2 + y^2)u_{xy} + (x^2 - y^2)u_{yy} = 0$, for region $x > 0, y > 0$.	K2
39	Find region in which partial differential equation $x^2 u_{xx} + x(y^2 - 1)u_{xy} + y(x^2 - y^2)u_{yy} = 0$, is hyperbolic?	K2
40	Use separation of variables method to solve following PDE: $\frac{\partial u}{\partial t} = \frac{\partial u}{\partial x}, u(0, x) = 2e^{-3x}.$	K3
41	Find the solution of initial value problem $u_t = u_{xx}, u(0, t) = 0, u(\pi, t) = 0$ and $u(x, 0) = \cos \cos x \sin \sin 5x.$	K2, K3
42	Find the temperature in a laterally insulated bar of length L whose ends are suddenly cooled at 0 degree Celsius and kept at that temperature, was initially at a uniform temperature u_0 .	K2, K3
43	Let $u(x, t)$ the solution of initial value problem $u_{tt} = u_{xx}, u(x, 0) = \cos \cos 5x$ and $u_t(x, 0) = 0$. Then find value of $u(1, 1)$?	K3
44	An elastic string of length l which is fastened at the ends $x = 0$ and $x = l$, is released from its horizontal position (zero initial displacement) with initial velocity $g(x)$ given as: $g(x) = \{x, 0 \leq x \leq \frac{l}{3}, \frac{l}{3} < x < l$ Find the displacement of the string at any instant of time.	K3
45	The boundary value problem governing the steady state temperature distribution in a flat, thin, rectangular plate of width a and insulated surface is given by $u(0, y) = 0, u(a, y) = 0, u(x, \infty) = 0, u(x, 0) = kx.$ Find steady state temperature in plate.	K3

STUDENT-CENTERED LEARNING (SELF-LEARNING TOWARDS LIFE-LONG-LEARNING)

A) SELF-LEARNING THROUGH MOOCs (Cognitive Skills): Certification